



VHIKK X



RETURN TO VHIKK X

ALGORITHM REFERENCE

Each combined position of the two toggle switches selects a different algorithm, giving 9 in total.

Below is an outline of the 9 modes indicating specific behaviour of the multi-modal controls.

The graphics on the left indicate toggle switch positions.

The firmware at launch is an extension/reimagining of the algorithms on the original Vhikk – So the general architecture consists of a set of oscillators [the **generator**] sent through a filter section which then passes into a delay-based structure [the **processor**] – with the specific architecture and character of these sections dependent on the selected mode.

Some of the algorithms implement what is referred to here as 'cluster scanning' – a smooth crossfading scan through sets of frequency/wave pairs. Imagine a wavetable scan parameter but each distinct wave position also has an associated frequency offset. In the modes implementing cluster scanning these wave/frequency offsets are different for every oscillator.

The processor in general consists of a delay-based structure which can be smoothly transformed from something resembling a delay [distinct [albeit processed] repeats] to something resembling a reverb [diffused repeats, increasing echo density, modulation].

There is a frequency shifter integrated into the feedback loop of the processor, with FEED determining the frequency shifter's mix type – [CCW: output direct from freq shift] [CW: freq shift output mixed with raw delay output for pronounced moving multi-notch-filter effect].

With the BASIS attenuverter at maximum, BASIS will track at 1V per octave.

For all of these algorithms, the VCA is placed after the filter section and before the processor.

The stereo audio inputs are mixed with the generator signal and routed through the filter/tone-shaping section and the processor.

The general parameter space is mapped out below, but some parameters have behaviour specific to individual algorithms [this is detailed in the next section].

[Note: CW: clockwise, CCW: counterclockwise]

WARP - Two different types of generator cross-mod - neutral position 12:00

SPAN - Oscillator spread/tuning - neutral position 12:00

MIX - Dry/wet blend between generator and processor - neutral position 12:00

FEED - Processor feedback - neutral position 12:00

MORPH - Generator timbral scanning

SEED - Manual internal randomisation scanning - delay taps, freq/wave offsets and notch spacing

SCAN - Randomisation scanning of internal timbral/waveform parameters

BASIS - Root frequency for all oscillators

TIME - Root scaling of delay lengths

FORM - Processor structure

FUSE - Algorithm-dependent timbral control

FIELD - Filtering centre frequency - neutral position 12:00

CELL - Frequency shift amount





- 2 oscillators [analogue-style] with dungeon metallic delay/reverb
 - WARP:
 - CCW: exponential cross-mod
 - CW: ring-mod level
 - SPAN: modulator relative tuning offset
 - MORPH: carrier waveform scan - pulse/saw/sine
 - FUSE:
 - CCW: chorus mix
 - CW: filter resonance and shape
 - SEED: multi-notch filter freq randomisation + delay tap spacing
 - SCAN: modulator waveform scan - pulse/saw/sine
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- dynamic waveshaping oscillator pair with refleakta diffuse delay structure
 - WARP:
 - CCW: modulation index + modulator feedback
 - CW: modulation index + carrier feedback
 - SPAN:
 - CCW: carrier frequency
 - CW: carrier frequency + stereo spread
 - MORPH: carrier waveform scan
 - FUSE:
 - CCW: delay line audio rate modulation
 - CW: saturation gain
 - SEED: multi-notch filter freq randomisation + delay tap spacing
 - SCAN: base waveshape set randomisation
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- multi-sine cluster oscillator bank with dungeon metallic delay/reverb
 - WARP:
 - CCW: oscillator phase feedback
 - CW: inter-oscillator exponential frequency cross-mod
 - SPAN:
 - CCW: frequency spread
 - CW: frequency + stereo spread
 - MORPH: cluster scan
 - FUSE:
 - CCW: chorus mix
 - CW: oscillator noise-modulation
 - SEED: cluster pitch offset randomisation + delay tap spacing
 - SCAN: inter-oscillator ring-mod mix
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- 4 oscillator wavetable scanning with dungeon metallic delay/reverb
 - WARP:
 - CCW: continuous oscillator bit depth reduction
 - CW: inter-oscillator exponential frequency cross-mod
 - SPAN:
 - CCW: frequency + waveform spread
 - CW: frequency + stereo spread
 - MORPH: wavetable scan
 - FUSE:
 - CCW: filter bandpass morph with parallel sine oscillator
 - CW: chorus mix and filter resonance + shape
 - SEED: pitch and wave offset randomisation + delay tap spacing
 - SCAN: base wavetable set randomisation



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- 4 oscillator wavetable scanning with dungeon metallic delay/reverb
 - WARP:
 - CCW: inter-oscillator phase modulation
 - CW: inter-oscillator exponential frequency cross-mod
 - SPAN:
 - CCW: frequency + waveform spread
 - CW: frequency + stereo spread
 - MORPH: wavetable scan
 - FUSE:
 - CCW: chorus mix
 - CW: chorus mix and filter resonance
 - SEED: pitch and wave offset randomisation + delay tap spacing
 - SCAN: base wavetable set randomisation





- 4 oscillator wavetable scanning with reflektas diffuse delay structure
 - WARP:
 - CCW: inter-oscillator phase modulation
 - CW: inter-oscillator exponential frequency cross-mod
 - SPAN:
 - CCW: frequency + waveform spread
 - CW: frequency + stereo spread
 - MORPH: wavetable scan
 - FUSE:
 - CCW: oscillator -> allpass+delay line audio rate modulation
 - CW: individual oscillator wavefold amount
 - SEED: pitch and wave offset randomisation + delay tap spacing
 - SCAN: base wavetable set randomisation
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- 4 oscillator diatonic-quantised cluster scanning with reflektas diffuse delay structure
 - WARP:
 - CCW: inter-oscillator phase modulation
 - CW: inter-oscillator exponential frequency cross-mod
 - SPAN: bipolar quantised cluster-frequency spread - CW introduces stereo spread
 - MORPH: cluster scan
 - FUSE:
 - CCW: stereo chorus mix
 - CW: saturation mix and gain
 - SEED: cluster pitch and wave offset randomisation + delay tap spacing
 - SCAN: base wavetable set randomisation
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- 4 oscillator cluster scanning with dungeon metallic delay/reverb
- WARP:
 - CCW: inter-oscillator phase modulation
 - CW: inter-oscillator exponential frequency cross-mod
- SPAN: bipolar cluster-frequency spread - CW introduces stereo spread
- MORPH: cluster scan
- FUSE:
 - CCW: stereo chorus mix
 - CW: saturation mix and gain
- SEED: cluster pitch and wave offset randomisation + delay tap spacing
- SCAN: base wavetable set randomisation



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- 4 oscillator cluster scanning with reflektä diffuse delay structure
 - WARP:
 - CCW: inter-oscillator phase modulation
 - CW: inter-oscillator exponential frequency cross-mod
 - SPAN: bipolar cluster-frequency spread - CW introduces stereo spread
 - MORPH: cluster scan
 - FUSE:
 - CCW: stereo chorus mix
 - CW: saturation mix and gain
 - SEED: cluster pitch and wave offset randomisation + delay tap spacing
 - SCAN: base wavetable set randomisation

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